Elizabeth Dietrich

University of California, Berkeley

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Position

University of California, Berkeley *Berkeley, USA*
Graduate Student Researcher *08/2023 – Today*
Electrical Engineering and Computer Sciences

Interests

Methods reachability analysis · formal methods · data-driven control
Topics autonomous systems · control theory · statistical learning

Education

University of California, Berkeley *Berkeley, CA, USA*
PhD in Electrical Engineering and Computer Sciences *08/2023 – Today*
Advisor: Murat Arcak

Indiana University, Bloomington *Bloomington, IN, USA*
B.S. in Computer Science, Minor in Mathematics *08/2017 – 05/2021*
Thesis: A beginner's guide to Iris, Coq and separation logic, Advisor: Dennis Shasha

Visits

Norwegian University of Science and Technology (NTNU) *Trondheim, Norway*
Visiting Researcher *06/2025 – 05/2026*
MC Lab (Asgeir J Sørensen and Roger Skjetne)

Norwegian University of Science and Technology (NTNU) *Trondheim, Norway*
Visiting Researcher *05/2024 – 07/2024*
MC Lab (Asgeir J Sørensen)

Publications

** first author(s)*

Preprints

[P1] Varun Madabushi*, **Elizabeth Dietrich***, Hanna Krasowski, and Maegan Tucker. *Finite-Step Invariant Sets for Hybrid Systems with Probabilistic Guarantees*. arXiv:2604.05102. 2026

Journal Articles

[J1] Hanna Krasowski*, **Elizabeth Dietrich***, Emir Cem Gezer, Roger Skjetne, Asgeir Johan Sørensen, and Murat Arcak. "pac-STL: PAC-Bounded Signal Temporal Logic from Data-Driven Reachability Analysis". In: *Accepted in IEEE Control Systems Letters* (2026)

Conference Papers

[C5] **Elizabeth Dietrich***, Hanna Krasowski*, and Murat Arcak. "Probably Approximately Correct (PAC) Guarantees for Data-Driven Reachability Analysis: A Theoretical and Empirical Comparison". In: *Accepted for Proc. of the IEEE Conference on Decision and Control (CDC)*. 2026

[C4] **Elizabeth Dietrich***, Hanna Krasowski, Vegard Flovik, and Murat Arcak. "Importance Sampling for Statistical Certification of Viable Initial Sets". In: *Accepted for Proc. of the IEEE Conference on Decision and Control (CDC)*. 2026

[C3] **Elizabeth Dietrich***, Rosalyn Devonport, Stephen Tu, and Murat Arcak. "Data-Driven Reachability with Scenario Optimization and the Holdout Method". In: *2025 IEEE 64th Conference on Decision and Control (CDC)*. 2025, pp. 3925–3931

[C2] **Elizabeth Dietrich***, Emir Cem Gezer*, Bingzhuo Zhong, Murat Arcak, Majid Zamani, Roger Skjetne, and Asgeir Johan Sørensen. "Symbolic Control for Autonomous Docking of Marine Surface Vessels". In: *2025 IEEE 64th Conference on Decision and Control (CDC)*. 2025, pp. 3021–3026

[C1] **Elizabeth Dietrich***, Alex Devonport, and Murat Arcak. "Nonconvex scenario optimization for data-driven reachability". In: *Proceedings of the 6th Annual Learning for Dynamics Control Conference*. Vol. 242. Proceedings of Machine Learning Research. PMLR, 15–17 Jul 2024, pp. 514–527

Invited Talks

2025 NSF Frontier Review, UC Berkeley · SFI Autoship Review, NTNU · Marine Cybernetics Seminar, NTNU · ANSR Seminar, UC Berkeley, online

2022 Informatics, Computing, and Engineering Seminar, IUB · Intelligent Systems Symposium, JHU APL

Mentoring

Student Projects

2025-2026 **Athul Krishnan (B.S.) & Chloe Yejin Lynn (B.S.)** *UCB*
NSF IRES Project on Autonomous Marine Operations for the Arctic

Student Theses

2025-2026 **Petter Dalhaug (M.S.)** *NTNU & UCB*
Safe Reinforcement Learning for Intelligent Ship Guidance and Control

Professional Experience

08/2021 – 07/2023 **Associate Staff: Research Scientist** · Johns Hopkins University Applied Physics Laboratory
05/2020 – 08/2021 **Undergraduate Research Assistant** · New York University, Courant Institute
05/2019 – 08/2019 **Software Engineer Intern** · Amazon

Teaching Experience

Spring 2025 **Optimization of Engineering Systems (MAS-E 238C),** Graduate Teaching Assistant (online)
Fall 2024 **Analysis and Control of Nonlinear Systems (MAS-E 250A),** Graduate Teaching Assistant (online)
Spring 2022 **MATLAB II,** STEM Academy Instructor JHUAPL
Fall 2020 **Training for Paraprofessionals in Student Affairs (EDUC-U208),** Undergraduate Teaching Assistant
Fall 2018 & 2019 **Introduction to Computer Science (CSCI-C211),** Undergraduate Teaching Assistant

Service to the Profession

Committees & Organizations

2024 - **EECS Equal Access to Application Assistance (EEAAA),** Assist underrepresented prospective PhD students on their graduate school applications to help equalize access to guidance on the application process. *UCB*

2026 **EECS PhD Visit Days Coordinator,** Coordinate visit days for close to 200 students admitted to the EECS PhD program. *UCB*

02/2026 **Application reviewer for Berkeley EECS MEng program,** Read and evaluate applications for admission to Berkeley's EECS professional master programs. *UCB*

03/2025 **EECS PhD CIR & CPSDA Visit Days ,** Organization of activities for prospective PhD students in controls, automation, learning, and embedded systems. *UCB*

Peer Review

Conferences (reviewed 4 submissions): Conference on Decision and Control (CDC)

Journals (reviewed 4 submissions): Control System Letters (L-CSS), Nonlinear Analysis: Hybrid Systems (NAHS)

Awards, Fellowships, & Grants

2026	**Graduate Diversity & Community Fellowship,** University of California, Berkeley
2026	**D-Lab Data Science & AI Fellowship,** University of California, Berkeley
2026	**EECS Departmental Chair's Graduate Award,** University of California, Berkeley
2025	**Fernström Fellowship,** University of California, Berkeley
2023	**NSF Graduate Research Fellowship (GRFP),** National Science Foundation
2021	**Kate Hevner Mueller Senior Award,** Indiana University, Bloomington
2020	**Goldwater Scholar,** Barry Goldwater Scholarship and Excellence in Education Foundation